

Final Report

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Abstract

This report analysed Singapore's Total Fertility Rate and Total Live-Births from 1960 to 2024 by ARIMA models. The period from 1960 to 2012 was used for model fitting and 2013-2024 was used for checking forecast accuracy. Different transformations are tested, including log transformation, first differencing and log differencing. The results show that log ARIMA models with first differencing are more suitable for both series. For TFR, the final model: $\log(\text{TFR})$ ARIMA(0,1,1)(1,0,0)[12]. For TLB, the final model: $\log(\text{TLB})$ ARIMA(0,1,1)(1,0,0)[12]. The models gave reasonable forecast for some testing period, but they do not follow the recent decline after 2021. This shows that ARIMA models are useful, but the forecast could be improved by adding additional fertility or demographic variables.

Introduction

Changes in fertility are a major demographic concern because they influence population growth, age structure and economic stability. We use Singapore birth and fertility data from 1960 to 2025, for this report, we focus only on Total Fertility Rate (TFR) and Total Live-Births (TLB) from 1960 to 2024. The dataset was obtained from Singapore Department of Statistics website. Total Fertility Rate (TFR) measures fertility levels, while Total Live-Births (TLB) measures the actual number of live births each year. Therefore, analysing both series gives a broader view of fertility change in Singapore

Singapore is an important case to study fertility because it has experienced a dramatic change over a long-term period. Fertility has fallen from a high level in the 1960 to a very low level today. Chen, Yip and Yap (2017) explained that Singapore's low fertility is strongly connected to marriage and first-birth behaviour, especially among single women aged 20-29 and married childless women aged 30-34. This is relevant for understanding the background of Singapore's fertility, as these behavioural changes may be reflected in today's TFR and TLB.

Recent official statistics also show that Singapore's fertility remains very low. The National Population and Talent Division (2025) reported that TFR was 0.97 in 2024, and that citizen births are declining over 2014 to 2024, with average of 32.9 thousand citizen births in 2015-2019 and with 30.4 thousand in 2020-2024. This is very important information to consider for forecasting that is based on previous years, because models that is trained on earlier data may not fully capture this recent change if the same trend was not present in the past.

The main goal of this report is to understand well time series models fitted to Singapore's Total Fertility Rate and Total Live-Births from 1960 to 2012 can forecast their changes from 2013 to 2024. The Methods section explains how the data was prepared, transformed and tested for stationarity before fitting ARIMA models. The Results section presents the main time series features, model comparison, residuals check and forecast accuracy. The Discussion section then considers the limitations of the selected models, especially their ability to accurately forecast the recent decline in TFR and TLB after 2013.

Methods

The analysis uses annual Singapore birth and fertility data from 1960 to 2024, and it is focused only on Total Fertility Rate (TFR) and Total Live-Births (TLB). First, data is cleaned to keep only required variable, TFR

and TLB, and rearranged in increasing year order. The period 1960-2012 is used as the training set for fitting the models, and for the testing set, 2013-2024 is kept for checking forecasting accuracy

Then, the original TFR and TLB series are plotted to understand the main time series trends. After this, several transformations are considered to check whether the series became more stable before fitting ARIMA models.

The transformations:

- the log transformation,

$$\log(X_t)$$

- the first difference,

$$(1 - B)X_t \quad \text{or} \quad \nabla X_t$$

- the log difference,

$$(1 - B)\log(X_t) \quad \text{or} \quad \nabla \log(X_t)$$

where X_t is either TFR or TLB

The KPSS test is used on training set to test stationarity, because ARIMA models usually require the series to be approximately stationary after transformations. In this test, stationarity is the assumption that is being tested. Therefore, when p-value > 0.05 , there is insufficient evidence to reject stationarity. When the p-value < 0.05 , the series is not stationary. The KPSS results are compared across the original and transformed series and are used with ACF/PACF plots and residual diagnostics to select model.

The transformed times series are examined by ACF and PACF plots to choose what type of ARIMA model is the most suitable to be fitted. The lags, shown as spikes in the plot, are considered to identify possible remaining dependence in the series. Then, candidate ARIMA models are fitted using the training set.

Residuals diagnostics are used to check whether the fitted models removed most of the time series dependence. Residual ACF and PACF plots are examined, and the residuals are expected to lie mostly within the confidence bounds and be close to white noise. Models that still have clear autocorrelation are not considered suitable for further forecasting. The fitted models are compared by AIC, where a lower AIC is better, but for choosing final model, the forecast accuracy is also needed to be considered.

Finally, the selected models are used to forecast for 2013-2024 period. The accuracy of the forecast is checked by Mean Squared Error (MSE) and Mean Absolute Error (MAE). Smaller MSE and MAE values indicate better forecast accuracy. Then, the testing set is plotted with the selected model for forecasting to compare the behavior of the actual data and forecasted values.

Results

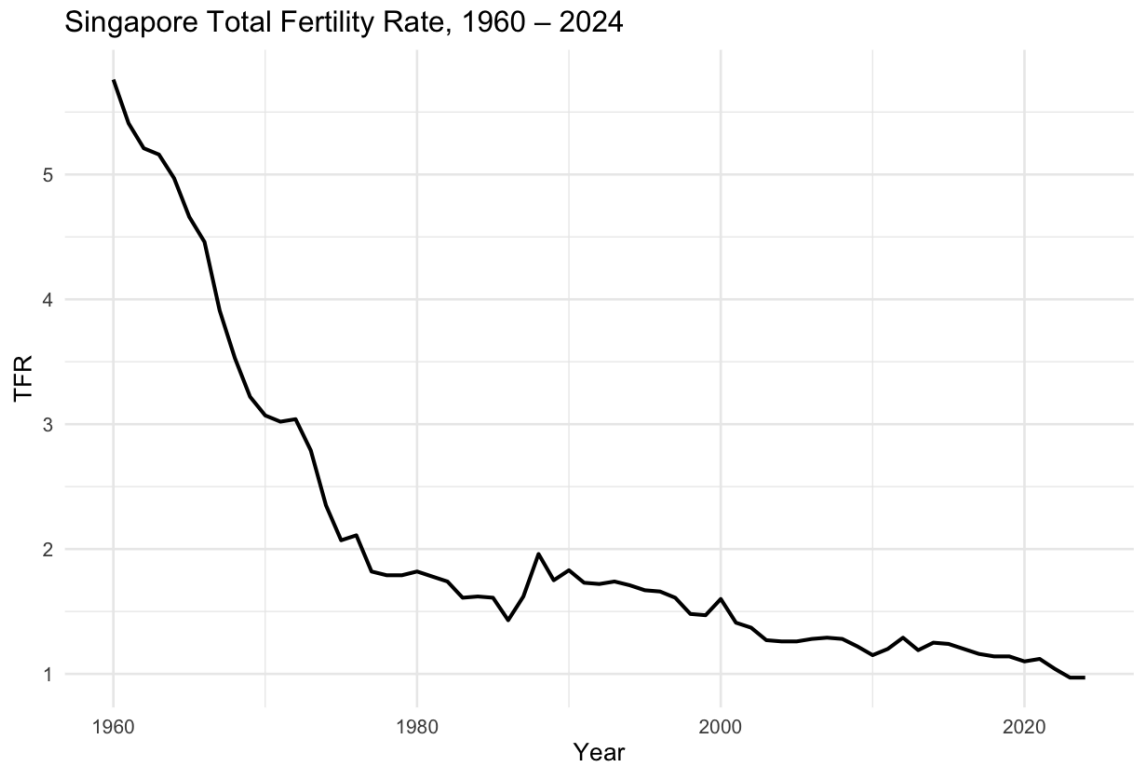


Figure 1: Singapore Annual Total Fertility Rate from 1960 to 2024

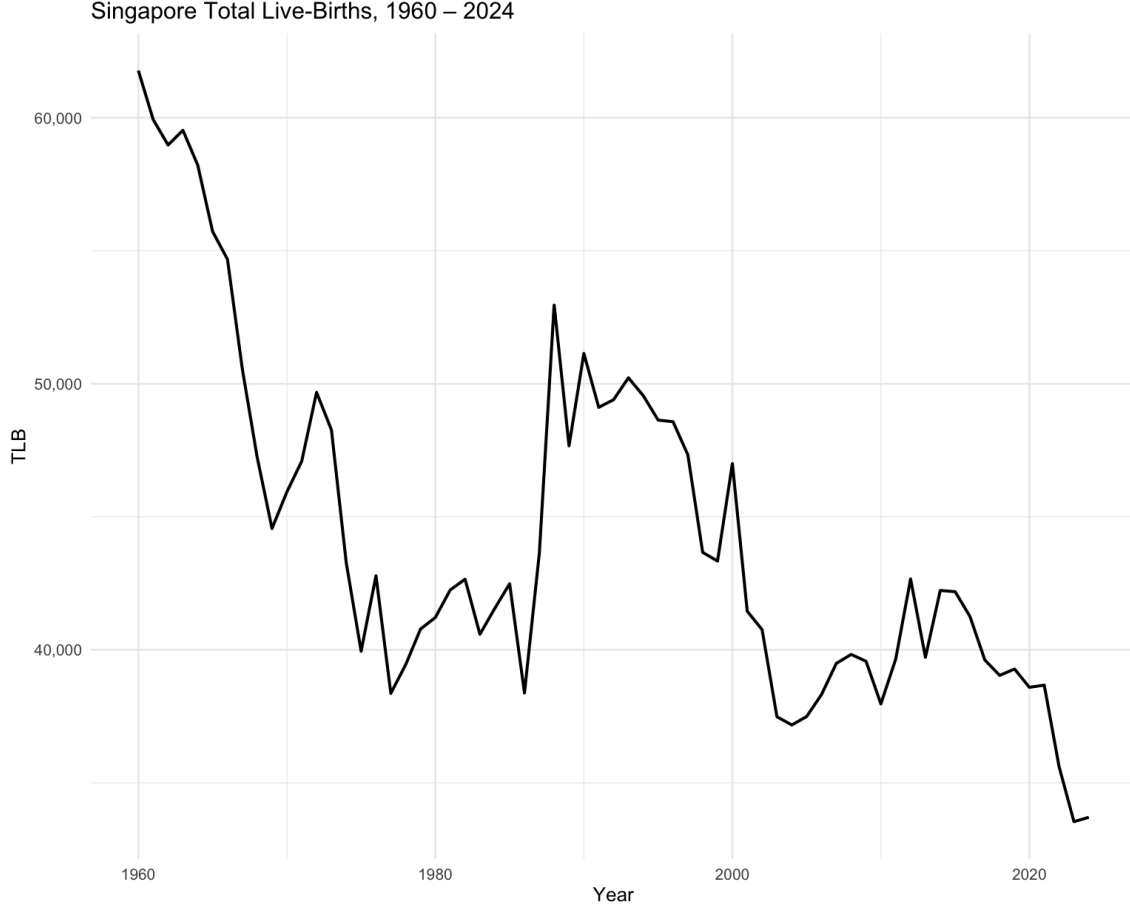


Figure 2: Singapore Annual Total Live-Births from 1960 to 2024

Figure 1 and Figure 2 show the original TFR and TLB from 1960 to 2024, respectively. The TFR series shows a clear long-term decline from a high level in 1960 to a much lower level in recent years. The TLB series also shows an overall decline over that period. There are some occasional rises, but in recent years it still drops to a very low number. These plots show that both series do not stay around one stable level, so we consider transformations before fitting ARIMA models.

Table 1: KPSS test results for TFR and TLB transformations, 1960 - 2012

Series	Transformation	kpss_stat	kpss_pvalue
TFR	X_t	1.0705934	0.0100000
TFR	$\log(X_t)$	1.2006081	0.0100000
TFR	$(1-B)X_t$	0.8412374	0.0100000
TFR	$(1-B)\log(X_t)$	0.5048485	0.0405747
TLB	X_t	0.6236379	0.0204875
TLB	$\log(X_t)$	0.6133656	0.0214213
TLB	$(1-B)X_t$	0.2126452	0.1000000
TLB	$(1-B)\log(X_t)$	0.1765042	0.1000000

Table 1 shows the KPSS test results for the original and transformed series. For TFR, all p-values are below 0.05. This means that TFR is more difficult to make stationary, however the log differenced series has the highest p-value among all TFR transformations, 0.041, so it is the closest to stationarity. For TLB, the

differenced and log differenced series have p-value of $0.1 > 0.05$, so that implies there is no strong evidence against stationarity. Therefore, differencing improves stationarity of TLB.

We check the ACF and PACF plots for the transformed training series, which are shown in Appendix A. There are noticeable spikes around lags 11-13. Because of this, we also consider ARIMA models with lag 11, lag 12 and lag 13 structures during model selection.

Then, we compare AIC of all these models and keep the models with the lowest or closest AIC values as our main candidates. The full AIC comparison tables are shown in Appendix B. For TFR, the lowest AIC is obtained by $\log(\text{TFR})$ ARIMA(0,1,1)(1,0,0)[12], followed closely by $\log(\text{TFR})$ ARIMA(1,1,1)(1,0,0)[12]. Other models have higher AIC values, so we do not keep them as our candidate models. For TLB, the lowest AIC is obtained by $\log(\text{TLB})$ ARIMA(0,1,1)(0,0,1)[13], followed by $\log(\text{TLB})$ ARIMA(0,1,1)(1,0,0)[12]. And as for TFR, the remaining models have higher AIC values, so we do not keep them as our candidates. Therefore, we keep these models as our main candidate models.

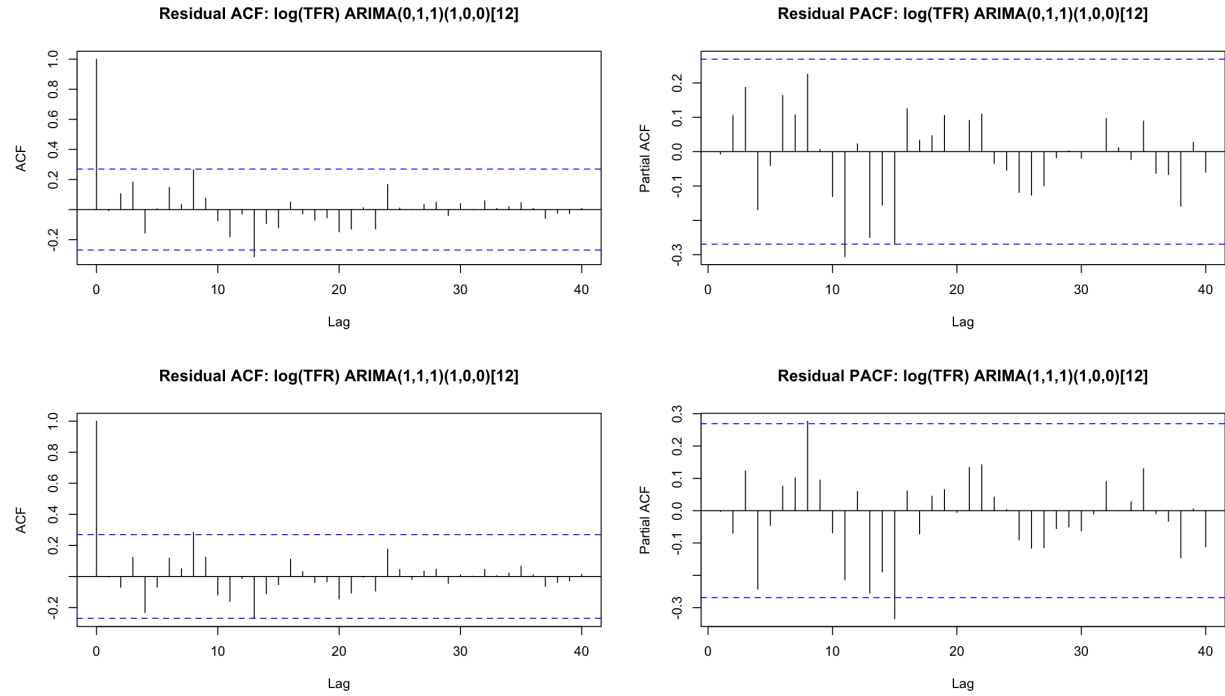


Figure 3: Residual ACF and PACF plots for TFR candidate models

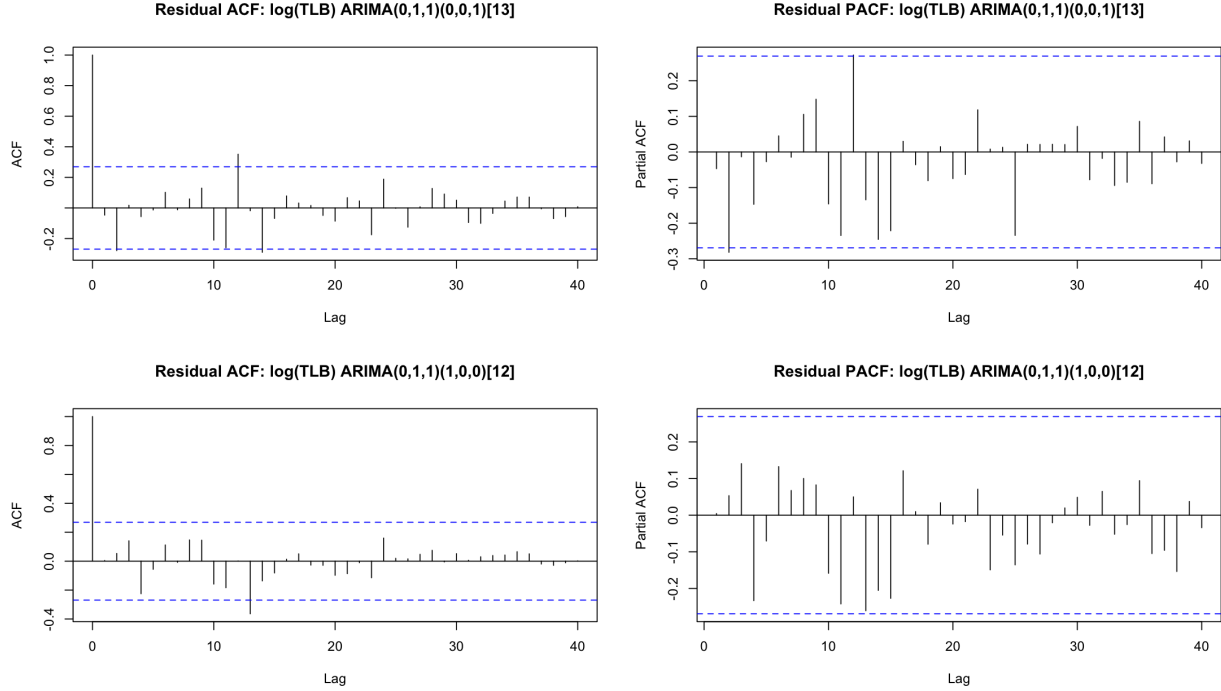


Figure 4: Residual ACF and PACF plots for TLB candidate models

After choosing the main candidate models using AIC, we check the residual ACF and PACF plots. Figure 3 and Figure 4 shows the residual ACF and PACF plots for the main candidate models. For the TFR candidate models, most residual spikes are inside the confidence bounds, but a few of them are outside the bounds. For the TLB candidate models, the result is similar. Most spikes are within the bounds and there are still a few spikes around some lags. Therefore, the residuals checks do not clearly show what model to choose, we keep these models and then check their forecast accuracy using MSE and MAE

Table 2: AIC and forecast accuracy comparison for main candidate models, 2013–2024

Series	Model	AIC	MSE	MAE
TFR	log(TFR) ARIMA(0,1,1)(1,0,0)[12]	-137.10	0.0063	0.0580
TFR	log(TFR) ARIMA(1,1,1)(1,0,0)[12]	-136.48	0.0090	0.0728
TLB	log(TLB) ARIMA(0,1,1)(0,0,1)[13]	-147.74	44,380,692	5,486.39
TLB	log(TLB) ARIMA(0,1,1)(1,0,0)[12]	-146.74	10,748,047	2,426.10

Table 2 shows the AIC, MSE and MAE values for the main candidate models. For TFR, log(TFR) ARIMA(0,1,1)(1,0,0)[12] has the lowest AIC, MSE and MAE. Therefore, we select this model as final TFR model. For TLB, log(TLB) ARIMA(0,1,1)(1,0,0)[12] has the lowest MSE and MAE, but it does not have the lowest AIC. However, we are forecasting TFR and TLB, we select this model as final TLB model because it gives better forecast accuracy.

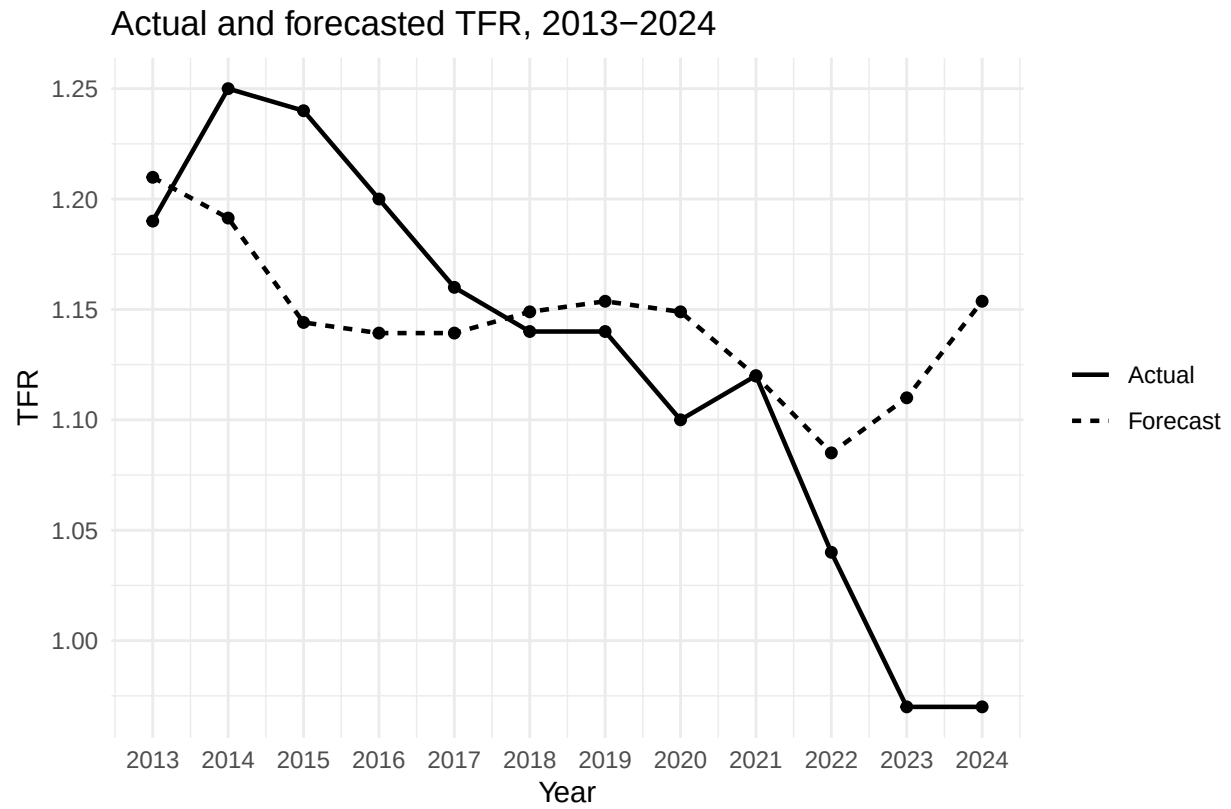


Figure 5: TFR Forecast Compared with Actual Values, 2013–2024

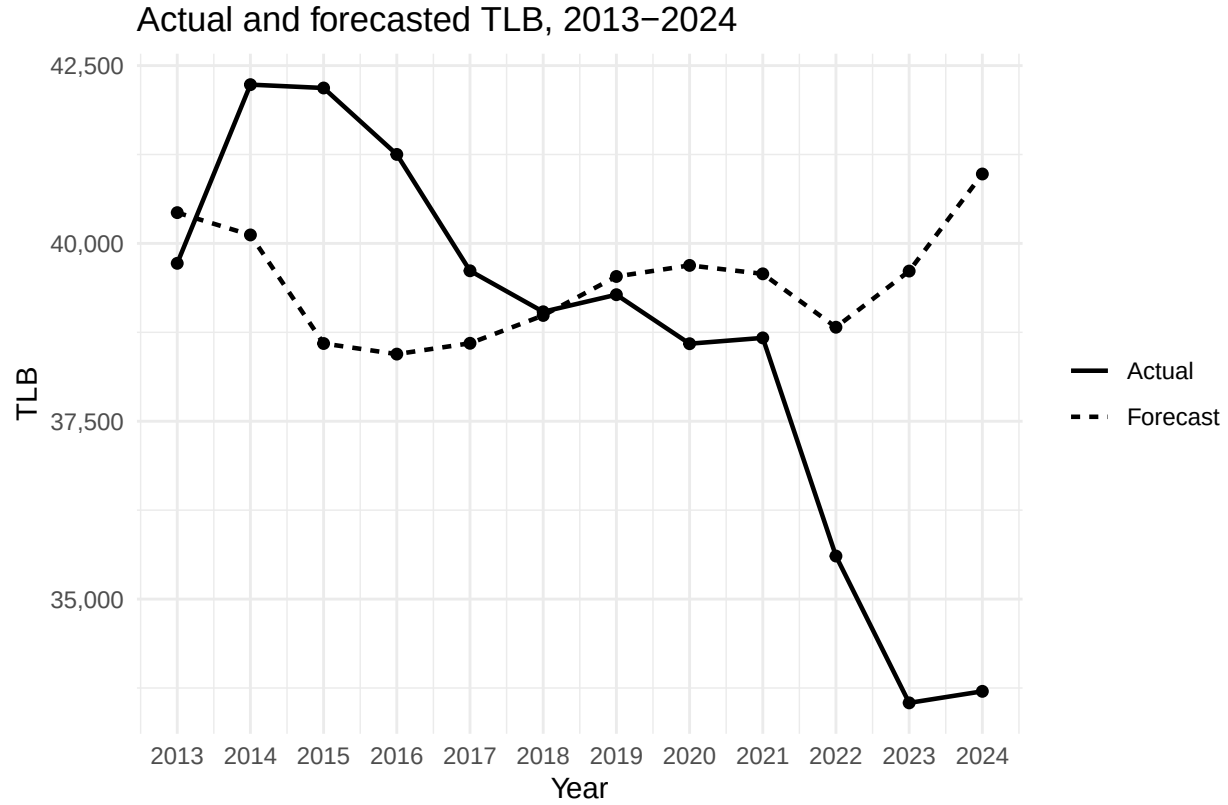


Figure 6: TLB Forecast Compared with Actual Values, 2013–2024

Figures 5 and 6 show the actual and forecasted values for the testing period from 2013 to 2024. The selected TFR model. Figure 5 shows that the selected TFR model captures the actual data trend at the beginning. However, after 2021, the actual TFR drops sharply, while the forecast only decreases slightly and then increases. Figure 6 shows that forecasted TLB has the same issues as TFR. The forecast follows the actual data before 2021, but after 2021 the actual data dramatically decreases, while the forecast rises. Overall, the models give reasonable forecasts for some period, but these models do not fully follow the recent decline in TFR and TLB

Discussion

The results show that the log ARIMA models are more suitable for TFR and TLB than the original models. The final models are also first differenced, so they are based on log differenced series.

For TFR, we select the final model because it has the lowest AIC, MSE and MAE among the candidate models. For TLB, the final model that we select does not have the lowest AIC, but it has much lower MSE and MAE. We selected it because we aim to forecast TLB, so we choose the model with better forecast accuracy.

However, both final models have a similar limitation. They give reasonable forecasts for some testing period, but they do not follow the decline trend after 2021 very well. This may happen because this decline trend appears in recent years and does not strongly present in the past data. Therefore, the forecasts are useful, but they should be adjusted for the most recent changes.

Overall, the models follow some of historical trend in the fertility data, but they are less accurate for the recent changes in fertility data. The forecast could be improved by using additional demographic variables.

Conclusion

This report analysed Singapore's Total Fertility Rate and Total Live-Births from 1960 to 2024 by ARIMA models. The period from 1960 to 2012 was used for model fitting and 2013-2024 was used for checking forecast accuracy. The results show that log ARIMA models with first differencing are more suitable for both series than the other transformations. For TFR, the final model that we selected was log(TFR) ARIMA(0,1,1)(1,0,0)[12], because this model has the lowest AIC, MSE and MAE. For TLB, we selected log(TLB) ARIMA(0,1,1)(1,0,0)[12] because it gave better forecast accuracy. In conclusion, the models gave reasonable forecast for some period over 2013-2024, but they do not follow the recent decline after 2021. This shows that ARIMA models are useful, but the forecast could be improved by adding additional fertility or demographic variables.

References

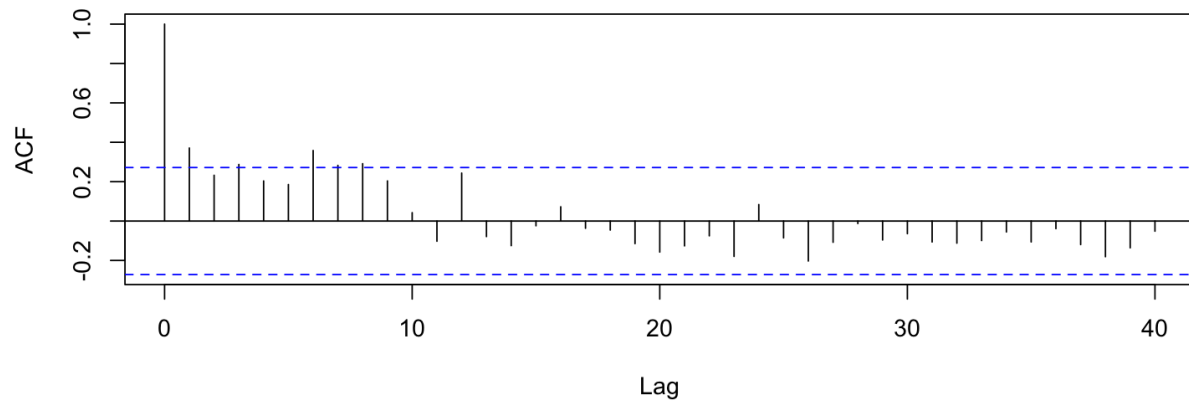
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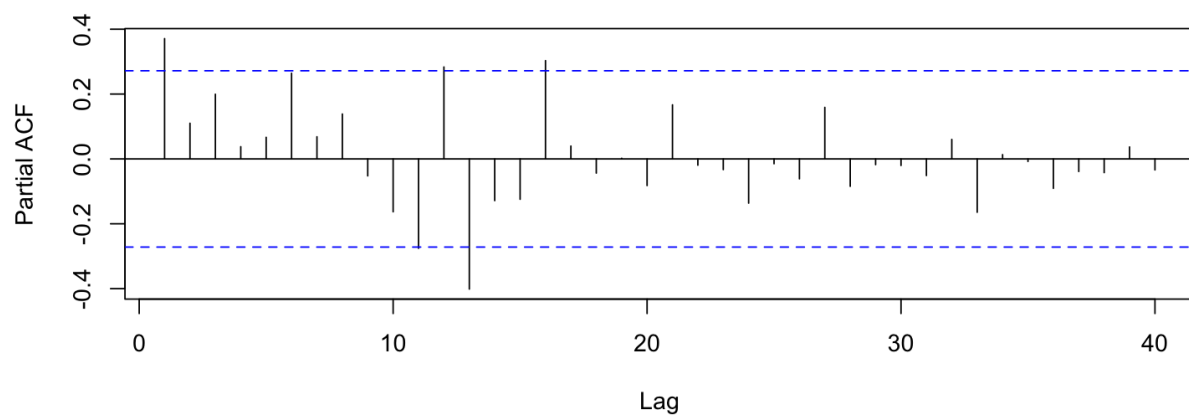
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Appendix A

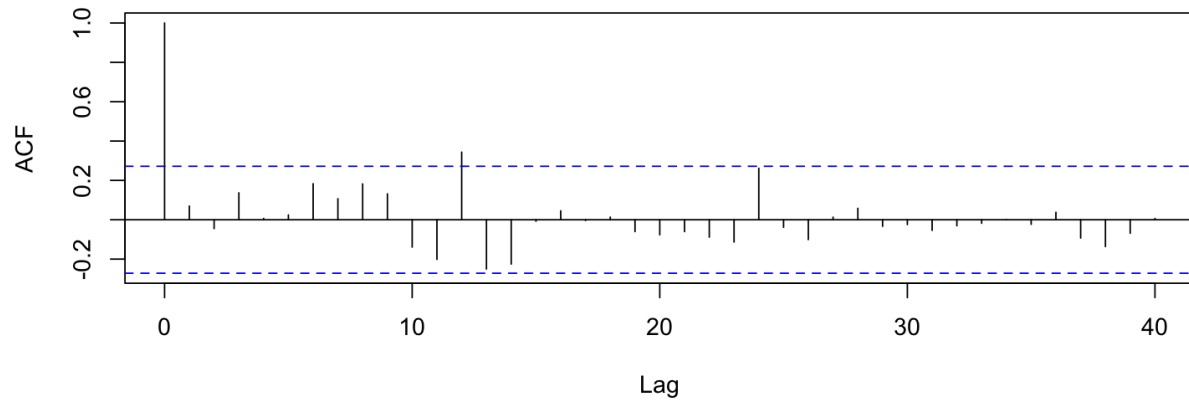
ACF of (1-B)TFR



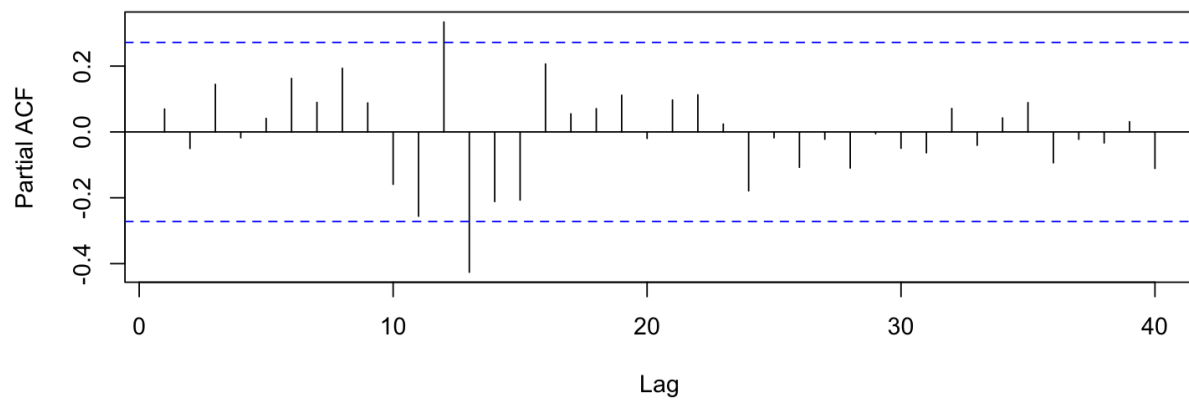
PACF of (1-B)TFR



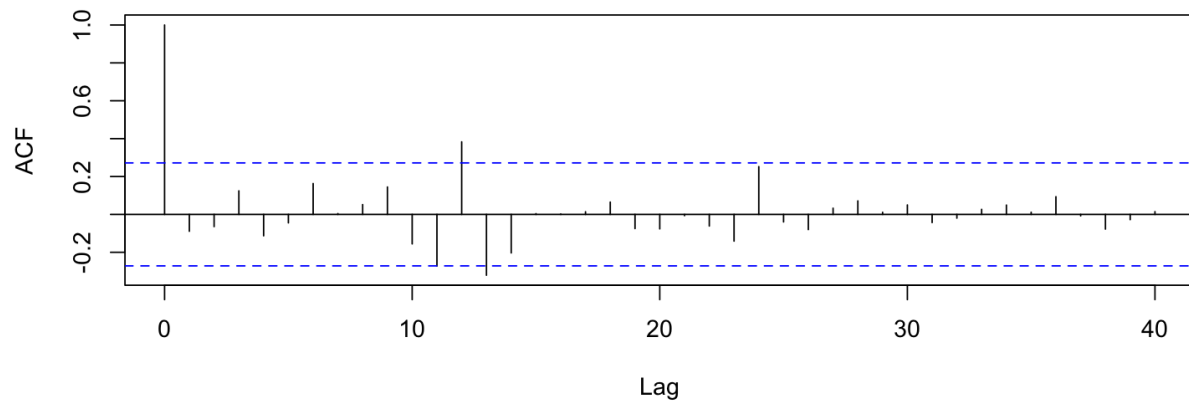
ACF of $(1-B)\log(\text{TFR})$



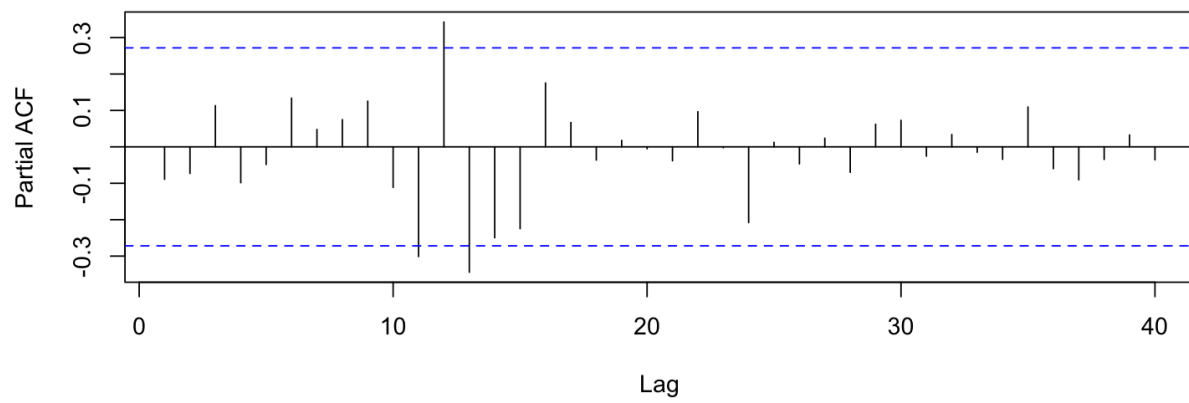
PACF of $(1-B)\log(\text{TFR})$

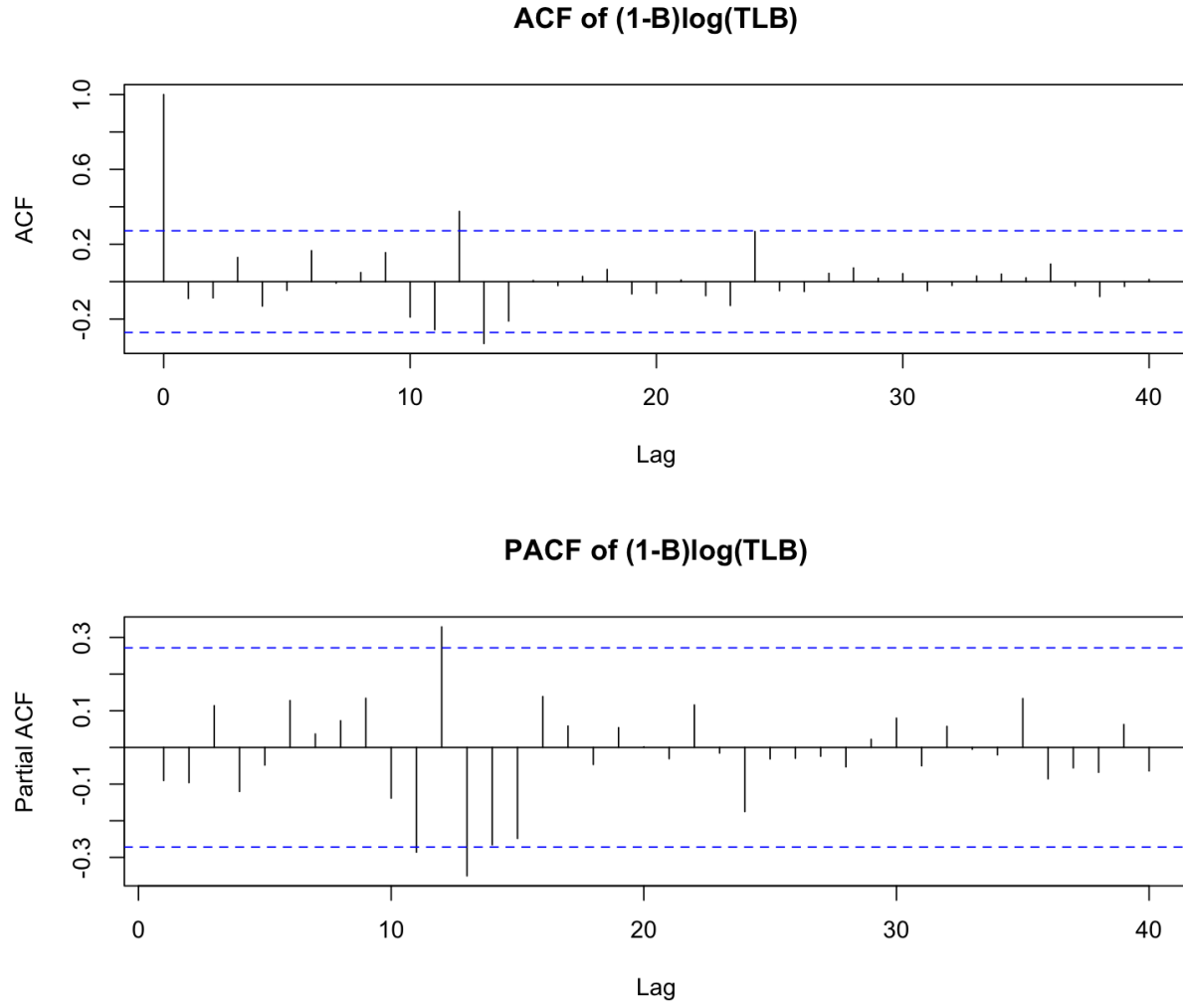


ACF of (1-B)TLB



PACF of (1-B)TLB





Appendix B

Table 3: AIC comparison for $\log(\text{TFR})$ ARIMA candidate models

Candidate model	AIC
$\log(\text{TFR})$ ARIMA(0,1,1)(1,0,0)[12]	-137.10
$\log(\text{TFR})$ ARIMA(1,1,1)(1,0,0)[12]	-136.48
$\log(\text{TFR})$ ARIMA(1,1,1)(0,0,1)[13]	-133.55
$\log(\text{TFR})$ ARIMA(0,1,1)(0,0,1)[12]	-131.83
$\log(\text{TFR})$ ARIMA(1,1,1)(0,0,1)[12]	-131.74
$\log(\text{TFR})$ ARIMA(1,1,1)(1,0,0)[13]	-130.17
$\log(\text{TFR})$ ARIMA(1,1,1)(0,0,1)[11]	-129.43
$\log(\text{TFR})$ ARIMA(1,1,1)(1,0,0)[11]	-128.59
$\log(\text{TFR})$ ARIMA(0,1,1)(0,0,1)[13]	-124.65
$\log(\text{TFR})$ ARIMA(0,1,1)(1,0,0)[13]	-124.28
$\log(\text{TFR})$ ARIMA(0,1,1)(0,0,1)[11]	-123.60
$\log(\text{TFR})$ ARIMA(0,1,1)(1,0,0)[11]	-123.57

Table 4: AIC comparison for log(TLB) ARIMA candidate models

Candidate model	AIC
log(TLB) ARIMA(0,1,1)(0,0,1)[13]	-147.74
log(TLB) ARIMA(0,1,1)(1,0,0)[12]	-146.74
log(TLB) ARIMA(1,1,1)(0,0,1)[13]	-146.20
log(TLB) ARIMA(1,1,1)(1,0,0)[12]	-145.08
log(TLB) ARIMA(0,1,1)(0,0,1)[12]	-143.38
log(TLB) ARIMA(0,1,1)(1,0,0)[13]	-143.33
log(TLB) ARIMA(1,1,1)(0,0,1)[12]	-141.51
log(TLB) ARIMA(0,1,1)(0,0,1)[11]	-141.13
log(TLB) ARIMA(0,1,1)(1,0,0)[11]	-140.25
log(TLB) ARIMA(1,1,1)(0,0,1)[11]	-139.26
log(TLB) ARIMA(1,1,1)(1,0,0)[11]	-138.37